J2 · ADAPTATION EXPERIMENT · 24 OCTOBER 2026

How Adaptation built a Sudoku game, twice

The same brief went through J2 twice. Once with today's Adaptation, and once with realization and certainty switched on. Both runs used the same model and the same 200-credit budget, one after the other. This document shows how each Adaptation turned the brief into workflows, how those workflows were built and checked, and what each run produced.

> **The brief, word for word:** "Build me a Sudoku game as a single web page. It's for casual players who play a quick game on their phone, like on a commute. It should feel friendlier than the cluttered Sudoku sites out there, and I want people to enjoy it enough to come back. Include a few difficulty levels and whatever helps a beginner without spoiling the puzzle. Keep it to one file, with no sign-up and no ads. You decide the details; tell me what you chose."

Flow 1 · Base J2 `J2_REALIZATION=0`

Adaptation groups the charter's items into the smallest set of workflows, in one call. Builders make their own choices. Problems are found by Audit after the build and fixed in later cycles.

Run `runs/base/j2-20261024-211320` · 21:13–21:32

Flow 2 · Realization × certainty `J2_REALIZATION=1`

Before proposing, Adaptation lists factors, cuts the work by kind of thinking, reads certainty, and scores every workflow with a check step, revising until it passes. It writes its decisions first, then proposes the build workflows to J2.

Run `runs/realization/j2-20261024-213253` · 21:33–22:14

RESULT	FLOW 1 · BASE	FLOW 2 · REALIZATION
Outcome	Done: workable, every done-when item passes	Done: workable, every done-when item passes
Time	`19 min (first proposal after 1 min)`	`42 min (first proposal after 15½ min of planning)`
AI calls / credits / cycles	`37 / 119 of 200 / 3`	`67 (11 planning) / 128 of 200 / 4`
Workflows planned before building	5 (by grouping items)	14 in the plan: 5 decisions, 9 build. 7 went into J2
First Audit	9 of 10 items pass; saved-game resume broken	**8 of 8 items pass**; nav condition open
Final game	"Sudoku Break", 61 KB, calm green	"Cozy Sudoku", 81 KB, warm orange
Puzzle quality (hidden test, 10 per level)	Always one solution; ~1 ms; Hard = fewer clues only	Always one solution; ~1 ms; Hard has fewer clues and is a little harder

One run each. AI output varies between runs, so treat this as a first look, not a result. Each run's Identity also wrote its own charter (10 items in Flow 1, 8 in Flow 2), so some differences come from Identity, not Adaptation.

Flow 1 · How base Adaptation made the workflows

BASE J2

Requester
BRIEF

The brief above. No follow-up questions were needed.

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Identity
J2.A · CHARTER

Charter: 10 done-when items
One HTML file · 9×9 grid with number pad · 3 levels, each with exactly one solution · notes, highlights, duplicate marks, undo · a hint that explains without filling in · win message · saved game and best times in the browser · no sign-up, ads or tracking · no overflow, 44px targets · no JS errors and working links

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Adaptation
J2.B · 1 CALL · ~1 MIN

W-1 Game board, number pad and difficulty
items 1–3 · 30 cr

W-2 Beginner help and hint
items 4–5 · 20 cr

W-3 Completion message and saved progress
items 6–7 · 20 cr

W-4 No sign-up, no ads promise
item 8 · 15 cr

W-5 Phone-first styling and navigation
items 9–10 · 10 cr

How it split the work
"The smallest set that covers all ten done-when items": done-when items grouped by screen area. No factors, no certainty reading, no decisions written down. How difficulty works, how hints behave and what gets saved were left for each builder to choose.

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Priority
J2.C

GRANT all five: 95 credits
Each priced at the rate-card minimum for its items. It flagged the game workflow (W-1) as the riskiest.

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Build → Audit
J2.E–H · 3 CYCLES

CYCLE 1

`W-1…W-5` the five planned workflows

Audit: 9/10. Item 7 fails: restoring a saved game redraws the board, but the game's own state still belongs to a new puzzle. N6 (no invented content) flags digits.

CYCLE 2

Adaptation asks for resume plus wider cells (24 cr). **Priority COUNTER:** funds resume only (8).

`W-6` Resume through the game's load step

Audit: all items pass. Only N6 remains, which Audit calls a false positive: they're timer and board digits.

CYCLE 3

`W-7` Pad digits drawn by CSS

`W-8` Timer starts as a word ("Ready")

Audit: every item and condition met. Done.
119 credits, 37 calls, 10 site versions.

Flow 2 · How Adaptation planned with realization and certainty

INSIDE ADAPTATION · 15½ MIN · 11 CALLS

Identity
J2.A · CHARTER

Charter: 8 done-when items
One HTML file, no sign-up or ads · grid playable by tapping · 3 levels, each a valid puzzle · notes, undo, highlights, optional mistake check, a hint that explains without filling in · warm, encouraging look with few controls · saved progress and a streak or games-completed counter · congratulation message · no JS errors, overflow or broken links

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1 · Factors
WHAT MATTERS

8 from the brief
mobile-first tap targets · warm encouraging look, few controls · one file · no sign-up or ads · hints never give the answer · progress on the device only · reasons to come back · several levels

5 the brief never mentioned
every puzzle has exactly one solution · levels must feel different · storage may be unavailable or corrupt · generation fast on a phone · warm palette stays readable

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2 · Stages
REALIZATION AXIS

product
users: scope, levels, help rules

design
experience: look, layout, wording

engineering
structure: one file, data model

build
making it

Not used
business, compliance, operations, growth: no factor or item owned them

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3 · Certainty
CERTAINTY AXIS

Goal reading: analyze.
Facts and patterns are known, but several valid choices are open (levels, hints, look, data), and a wrong pick would waste the build. So: decide first, then build. Decide workflows read `analyze`; build workflows that follow them read `operate`. Nothing was read as `experiment`: "come back" became features (a streak and counter), not a probe.

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4 · Check step
SCORE · REVISE

ROUND 0 · 11 WORKFLOWS	0.72
Look, layout and wording · *two kinds of thinking*	0.72
Beginner helps · *vague done-test*	0.77
9 others	0.85–0.95

ROUND 1 · 13 WORKFLOWS	0.62
Split: Look and layout + Encouraging wording	SPLIT
Split: Beginner helps + Hint button	SPLIT
Hint button · *done-test*	0.75
Responsive layout · *inputs*	0.62

ROUND 2 · 14 WORKFLOWS	0.85 ✓
Hint button checks rewritten	0.95
Split: How to play section	SPLIT
Responsive layout	0.95
Every workflow ≥ 0.80	PASSES

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5 · Decisions
WRITTEN FIRST

Scope and return-visit
Easy 38–45 clues, simple techniques; Medium 30–36; Hard 24–29; streak and counter

Hint and help rules
one marked cell, explained, never a value

Look and layout
warm palette, contrast computed, cells can't be 44px at 360px

Encouraging wording
name "Cozy Sudoku", every message ≤ 12 words

Architecture and data model
file order, ids, state, storage

Two build workflows, "Beginner helps" and "How to play", delivered no done-when item, so a bug in our code wrote them up as decisions instead of building them. Their features still appeared, because the builders that covered those items followed them.

Flow 2 · The judged plan, then building in J2

REALIZATION × CERTAINTY

STAGE	DECIDE FIRST ANALYZE	BUILD OPERATE
product	Scope and return-visit features · Hint and help rules	·
design	Look and layout · Encouraging wording	·
engineering	Single-file architecture and data model	·
build	·	Page shell and warm look (1, 5) · Playable grid and number pad (2) · Puzzle generator and levels (3) · Hint button (4) · Saved progress and counters (6) · Completion celebration (7) · Responsive layout and links (8) · Beginner helps, How to play (not built as workflows: bug)

Numbers in brackets are the done-when items each build workflow delivers. Every item has exactly one deliverer; the decisions inform the items they shape.

Adaptation
J2 PROPOSAL

7 build workflows proposed, names and items taken from the plan
J2's validator dropped 2 (the generator and the hint) over their check fields. Adaptation re-proposed them in the same cycle, plus a nav revision.

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Priority
J2.C

GRANT 5 (60 cr), then GRANT the generator and hint (20 cr)
It cut the nav revision, because a "fix" needs an Audit finding and none existed yet. Adaptation accepted and argued a nav hurts a one-screen game.

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Build → Audit
J2.E–H · 4 CYCLES

CYCLE 1

`W-1…W-7` the seven planned workflows

Audit: 8/8 items pass. Conditions open: N8 (no nav or one main action) and N6 (digits).

CYCLE 2

`W-8` Resume: restoreGame was never defined

`W-9` Nav, one main action, plain wording

Audit: all pass; resume fixed. Only N6 remains.

CYCLE 3

`W-10` A race on load could wipe a saved game

`W-11` Counters written by script

Audit: all pass. N6 still flags the number-pad digits.

CYCLE 4

`W-12` Number-pad digits drawn by script

Audit: every item and condition met. Done. 128 credits, 67 calls, 15 site versions.

N6 ("no invented content") flagged ordinary Sudoku digits in both flows. It cost each run about one cycle that had nothing to do with Adaptation.

The two flows side by side

WHAT THE AXES CHANGED

STEP	FLOW 1 · BASE	FLOW 2 · REALIZATION × CERTAINTY
What Adaptation looks at first	The done-when list	Factors: 8 from the brief and 5 it added itself (uniqueness, levels that feel different, broken storage, speed, readable colours)
How the work is cut	By screen area, smallest set: 5 workflows	By kind of thinking (product, design, engineering, build): 5 decide + 9 build
Reading of uncertainty	None	analyze: decide first, then build. Per workflow: decide = analyze, build = operate
Check before building	Shape and price checks only	Check step: 2 revision rounds, 0.72 → 0.62 → 0.85. Split three mixed workflows; rewrote checks
Choices written down	None; each builder chose	5 decisions: levels, hint rules, palette and contrast, wording, data model. Assumptions marked
Time to first proposal	~1 min, 1 call	15½ min, 11 calls
First Audit	9/10; resume broken	8/8; nav condition open
Rework	3 fixes over 2 cycles: resume, N6 ×2	5 fixes over 3 cycles: resume ×2, nav, N6 ×2
Total	3 cycles · 37 calls · 119 cr · 19 min	4 cycles · 67 calls · 128 cr · 42 min

What realization and certainty changed

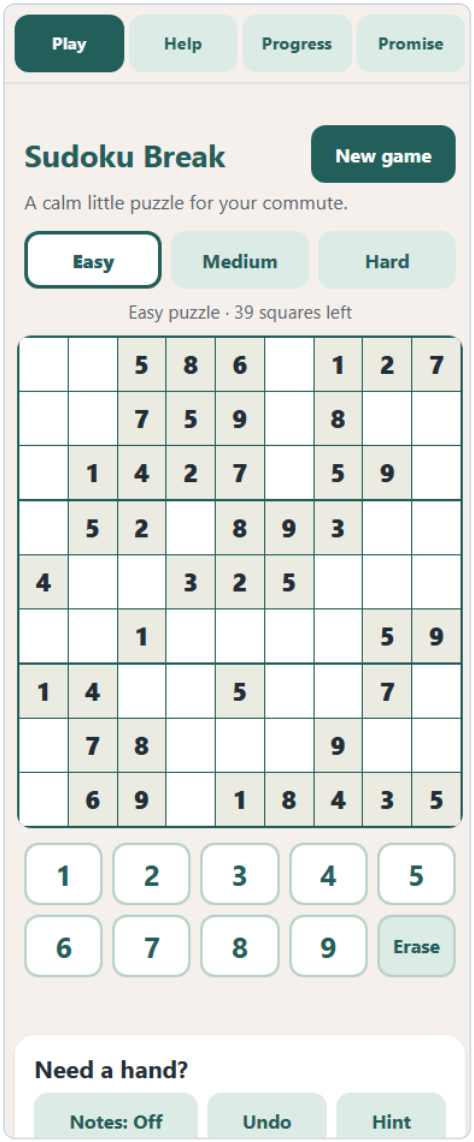
- Quality factors nobody asked for were planned for from the start.
- Choices were made once and written down, not guessed by each builder.
- The check step split work that mixed two kinds of thinking.
- Trade-offs were found before building: 44px cells can't fit at 360px, and technique-based difficulty only for Easy.
- The first build passed every done-when item.

What stayed the same

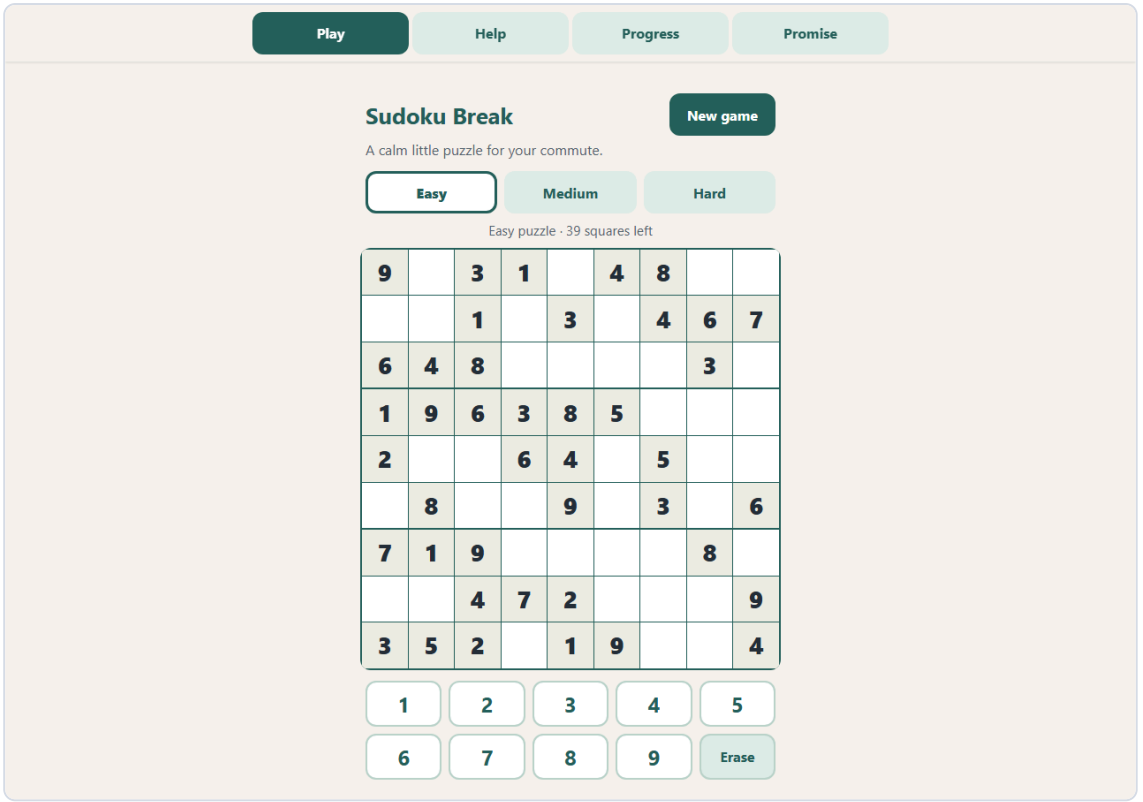
- Identity's charter still sets the features. Adaptation adds none.
- Integration bugs between builders still happened (a function nobody defined, a race on load), despite the written data model.
- J2's own checks (the N6 false positive) set the finishing pace in both flows.
- Puzzle quality ended up about the same. Base also guaranteed one solution without being told.

HIDDEN TEST: 10 PUZZLES PER LEVEL THROUGH EACH GAME'S OWN GENERATOR	BASE: CLUES · UNIQUE · SIMPLE TECHNIQUES SOLVE IT	REALIZATION: CLUES · UNIQUE · SIMPLE TECHNIQUES SOLVE IT
Easy	42 · 10/10 · 10/10	38–45 · 10/10 · 10/10
Medium	34 · 10/10 · 10/10	30–36 · 10/10 · 9/10
Hard	28 · 10/10 · 9/10	24–29 · 10/10 · 8/10
Time for a new puzzle	~1 ms	~1 ms

Final output · Flow 1 · "Sudoku Break"



Phone, 390px wide, top of the page (it continues with the promise and progress panels)



Desktop, 1280 × 900

What the player gets

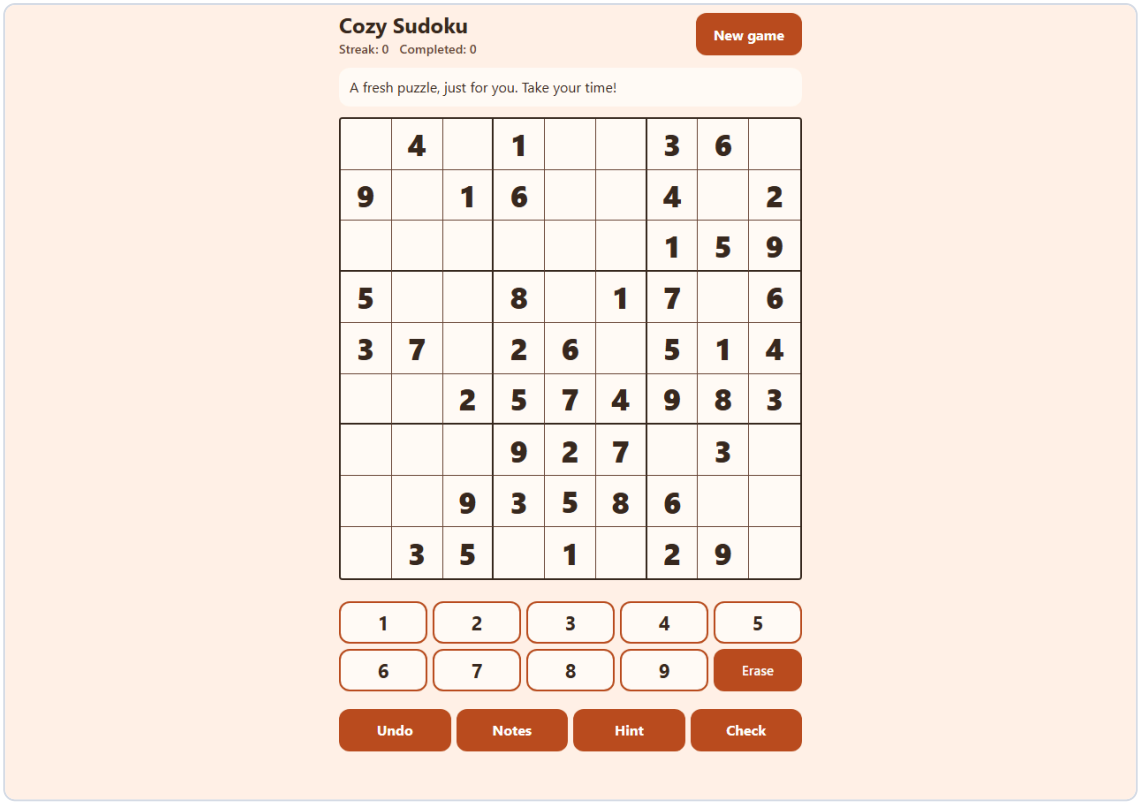
- A calm green look with a 4-tab nav: Play, Help, Progress, Promise.
- Easy, Medium and Hard above the grid; a 5 × 2 number pad with Erase.
- "Need a hand?" sits below the grid: Notes, Undo and a Hint that points to a square and explains why.
- An "Our promise to you" panel (no sign-up, no ads, no tracking). Further down, "Your progress": a timer, a saved game and best times per level.

Final output · Flow 2 · "Cozy Sudoku"

REALIZATION × CERTAINTY · SITE V15 · 81 KB



Phone, 390px wide, the whole page



Desktop, 1280 × 900

What the player gets

- A warm orange look, a streak and games-completed counter in the header, and a changing encouraging message ("A fresh puzzle, just for you. Take your time!").
- A one-screen game: grid, number pad with Erase, then Undo, Notes, Hint and Check, all within thumb reach. Levels and a folded "How to play" sit below.
- The hint marks one square and explains why from fixed templates, never a number. An optional mistake check.
- The game saves on the device and resumes on return. Wording, palette and levels follow the decisions written before building.

Screenshots taken in headless Chrome from each run's final `site/index.html`, at a fresh start (a new Easy puzzle, empty counters).